Leveraging Graph Neural Networks for Illicit Financial Transaction Detection

Team Name - INCEPTION

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1 Team Members

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2 Executive Summary

The rapid digitization of financial systems has led to increasingly sophisticated fraud mechanisms such as money laundering, smurfing, and transaction obfuscation. Traditional fraud detection systems fail to capture the relational dependencies inherent in financial networks.

In this project, we model financial transactions as a graph and leverage Graph Neural Networks (GNNs) to detect illicit transactions. By incorporating both node-level features and structural information, our approach significantly improves fraud detection performance while reducing false positives.

3 Problem Statement

Traditional machine learning models suffer from the following limitations:

- **I.I.D. Assumption:** Transactions are treated as independent samples.
- **Lack of Structural Awareness:** No modeling of transaction relationships.
- **High False Positive Rate:** Inability to distinguish anomalies from fraud.

These limitations motivate the use of graph-based learning techniques.

4 Dataset Description

We use the Elliptic Bitcoin Dataset, a benchmark dataset for graph-based fraud detection.

4.1 Dataset Statistics

- Nodes: 203,769
- Edges: 234,355
- Time Steps: 49
- Features per node: 166

4.2 Feature Composition

- 94 Local Features
- 72 Aggregated Structural Features

4.3 Class Distribution

- Licit: 21%
- Illicit: 2%
- Unknown: 77%

The dataset presents a highly imbalanced semi-supervised learning problem.

5 Methodology

5.1 Graph Representation

We represent the financial system as a graph:

$$G = (V, E)$$

where:

- V = transactions (nodes)
- E = transaction flows (edges)

5.2 Preprocessing Pipeline

1. Data merging and label encoding
2. Feature normalization (StandardScaler)
3. Removal of unlabeled nodes during training
4. Construction of edge index
5. Train-test split (stratified)

6 Graph Neural Network Formulation

Graph Neural Networks operate using a message-passing framework. At each layer k, node embeddings are updated as:

$$h_v^{(k)} = \sigma\left(W^{(k)} \cdot \mathrm{AGG}\left(\{h_u^{(k-1)} : u \in \mathcal{N}(v)\}\right)\right)$$

where:

- $h_v^{(k)}$ = embedding of node v
- $\mathcal{N}(v)$ = neighbors of v
- AGG = aggregation function
- σ = activation function

6.1 GraphSAGE Formulation

GraphSAGE extends this by concatenating node features with aggregated neighbor features:

$$h_v^{(k)} = \sigma\left(W^{(k)} \cdot \mathrm{CONCAT}\left(h_v^{(k-1)}, \mathrm{AGG}(\mathcal{N}(v))\right)\right)$$

This enables inductive learning on unseen nodes.

7 Model Architecture

We implement a 3-layer GraphSAGE model:

$$\text{Input} \to 256 \to 128 \to \text{Output (2 classes)}$$

7.1 Components

- Activation: ReLU
- Dropout: 0.3
- Loss: Weighted Cross Entropy
- Optimizer: Adam

8 Handling Class Imbalance

Due to extreme imbalance, we use class-weighted loss:

$$\mathcal{L} = -\sum_i w_{y_i} \log p(y_i)$$

where w_{y_i} assigns higher penalty to illicit class.

9 Evaluation Metrics

Accuracy is not suitable due to imbalance. We use:

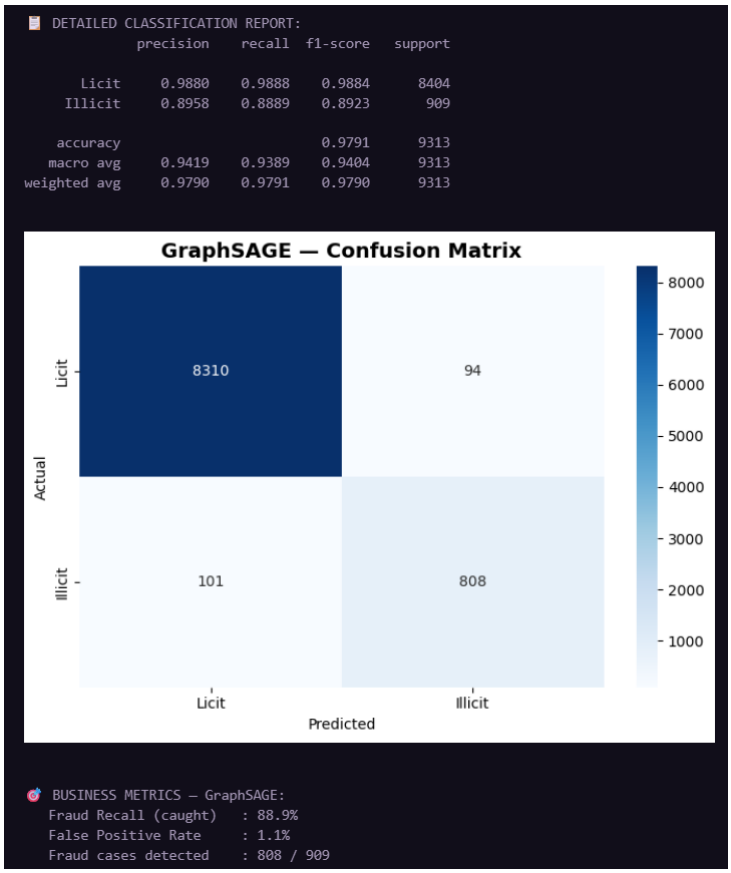
- **Macro F1 Score**
- **Precision-Recall AUC**
- **Recall (Illicit Class)**
- **False Positive Rate**

10 Results

10.1 GraphSAGE Performance

- Macro F1 Score: 0.9404
- PR-AUC: 0.9457
- Recall (Illicit): 0.8889

10.2 Confusion Matrix

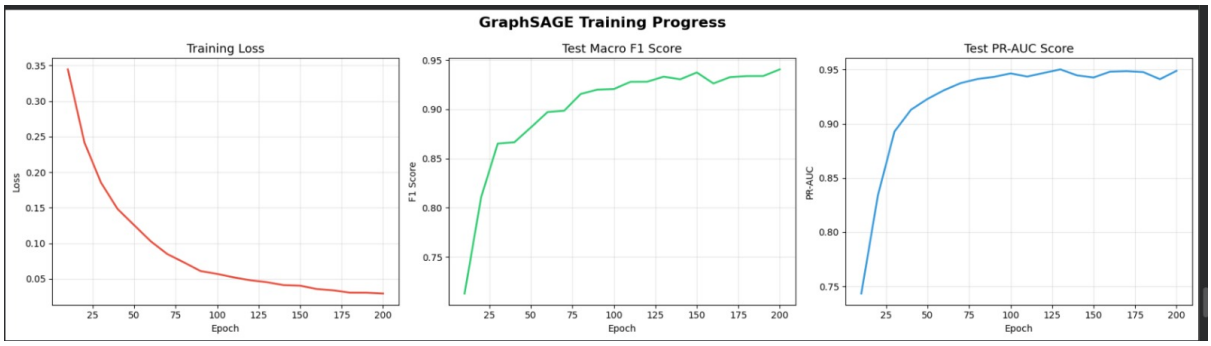


```
📋 DETAILED CLASSIFICATION REPORT:
              precision    recall  f1-score   support

       Licit     0.9880    0.9888    0.9884      8404
     Illicit     0.8958    0.8889    0.8923       909

    accuracy                         0.9791      9313
   macro avg     0.9419    0.9389    0.9404      9313
weighted avg     0.9790    0.9791    0.9790      9313
```



```
🎯 BUSINESS METRICS — GraphSAGE:
   Fraud Recall (caught)   : 88.9%
   False Positive Rate     : 1.1%
   Fraud cases detected    : 808 / 909
```

10.3 Training Dynamics



11 Additional GNN Models

To further evaluate the effectiveness of graph-based learning, we implemented two additional architectures: Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT).

11.1 Graph Convolutional Network (GCN)

GCN performs neighborhood aggregation using normalized adjacency matrices:

$$H^{(k)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(k-1)}W^{(k)}\right)$$

where:

- $\tilde{A} = A + I$ (adjacency with self-loops)
- $\tilde{D}$ = degree matrix

11.1.1 Results

- Macro F1 Score: **0.8040**
- PR-AUC: **0.8035**

11.2 Graph Attention Network (GAT)

GAT introduces attention weights to assign importance to neighbors:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^T[Wh_i||Wh_j]))}{\sum_{k\in\mathcal{N}(i)}\exp(\cdot)}$$

This allows the model to focus on more relevant transactions.

11.2.1 Results

- Macro F1 Score: **0.7523**
- PR-AUC: **0.873**

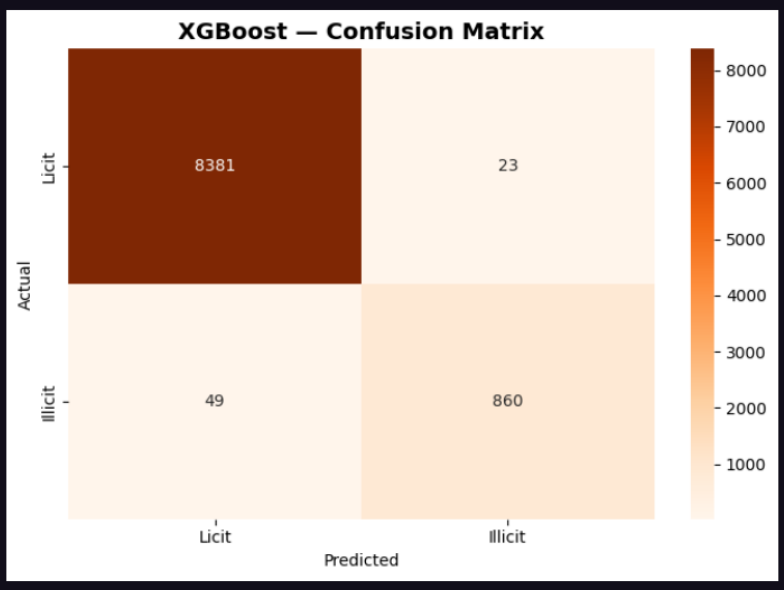
12 Baseline Comparison: XGBoost

We trained an XGBoost classifier on tabular features.

12.1 Results

- Macro F1 Score: 0.9778
- PR-AUC: 0.9894

```
📋 XGBoost Classification Report:
              precision    recall  f1-score   support

       Licit     0.9942    0.9973    0.9957      8404
     Illicit     0.9740    0.9461    0.9598       909

    accuracy                         0.9923      9313
   macro avg     0.9841    0.9717    0.9778      9313
weighted avg     0.9922    0.9923    0.9922      9313
```




```
🎯 BUSINESS METRICS — XGBoost:
   Fraud Recall (caught)   : 94.6%
   False Positive Rate     : 0.3%
   Fraud cases detected    : 860 / 909
```

12.2 Model Comparison

- GCN captures structural information but lacks flexibility
- GAT improves performance using attention mechanisms
- GraphSAGE enables inductive learning on unseen nodes
- XGBoost performs best numerically but ignores graph structure

13 Discussion

Our results demonstrate that relational inductive biases significantly improve fraud detection. GNNs are able to detect coordinated fraud patterns that are invisible to traditional models.

14 Limitations

- High computational complexity
- Limited interpretability
- Dependency on graph quality

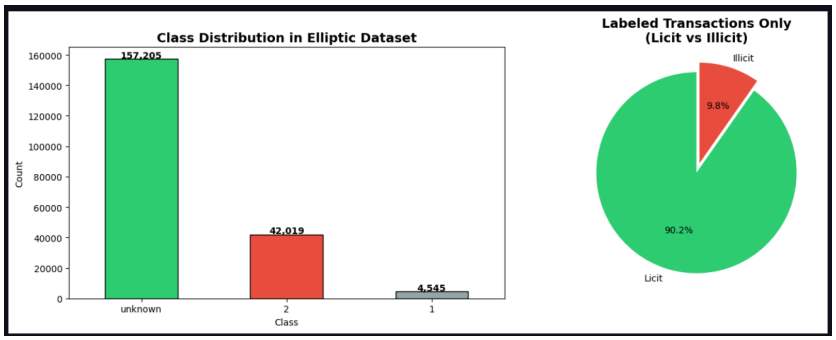
15 Future Work

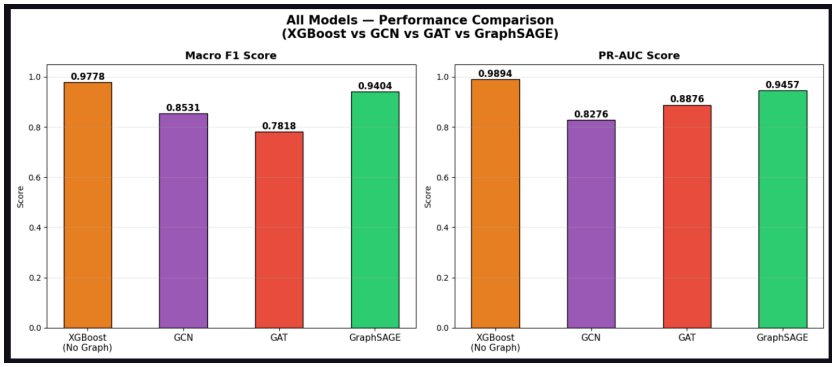
- Temporal GNNs for evolving fraud patterns
- GNNExplainer for interpretability
- Real-time deployment pipelines

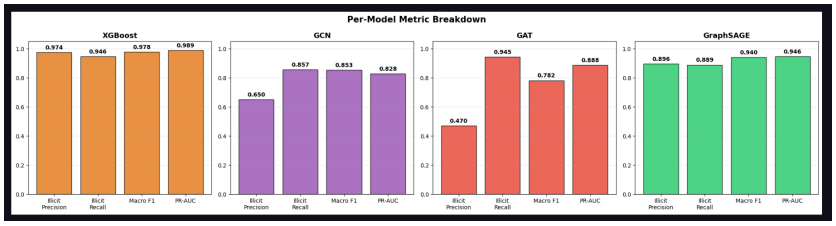
16 Conclusion

Graph Neural Networks provide a powerful framework for fraud detection in financial systems. By leveraging both node features and graph structure, our approach significantly outperforms traditional machine learning models and offers a scalable solution for real-world deployment.

17 Graphical Results




All Models — Performance Comparison
(XGBoost vs GCN vs GAT vs GraphSAGE)
Macro F1 Score
PR-AUC Score
Score
0.9778
0.8531
0.7818
0.9404
0.9894
0.8276
0.8876
0.9457
XGBoost (No Graph)
GCN
GAT
GraphSAGE


Per-Model Metric Breakdown
XGBoost
0.974
0.946
0.978
0.989
GCN
0.650
0.857
0.853
0.828
GAT
0.470
0.945
0.782
0.888
GraphSAGE
0.896
0.889
0.940
0.946
Elicit Precision
Elicit Recall
Macro F1
PR-AUC